

PROGRAM 4 - Create a dockerized application and push it to GitHub using Personal Access Token (PAT).

Steps

1. Create project directory

```
mkdir lab4 && cd lab4
```

2. Create files

- app.py
- requirements.txt
- Dockerfile

3. Create Dockerfile

Create Dockerfile:nano Dockerfile

```
FROM python:latest
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
EXPOSE 5000
CMD ["python", "app.py"]
```

3. Build Docker image

```
docker build -t program4 .
```

4. Run Docker container

```
docker run -p 5000:5000 program4
```

5. Verify output

Open browser: <http://localhost:5000>

6. Initialize Git

```
git init
```

```
git branch -M main
```

```
git add .
```

```
git commit -m "initial commit"
```

7. Push to GitHub using PAT

```
git remote add origin https://github.com/<username>/program4.git
```

```
git push -u origin main
```

OUTPUT

Dockerized application created and pushed to GitHub successfully.

The screenshot shows a GitHub repository named 'Prog4' by user 'ganeshvissapragada'. The repository is public and has 0 stars, 0 forks, and 0 watchers. The repository contains a README.md file, a Dockerfile, a package.json, a package-lock.json, a .gitignore, and a server.js file. The README.md file is titled 'lab4'.

File	Commit	Time
node_modules	first	2 months ago
.gitignore	first	2 months ago
Dockerfile	first	2 months ago
README.md	Update project name from lab3 to...	1 minute ago
package-lock.json	first	2 months ago
package.json	first	2 months ago
server.js	first	2 months ago

README

lab4